Introduction to Internet-of-Things (IoT)

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1. What is IoT?
2. Understanding IoT systems
3. Practical

1. What is IoT?

Learning objectives

By the end of this week, students should be able to:

- Explain the meaning of IoT
- Identify the main components of an IoT system
- Describe the basic IoT data flow
- Recognize IoT applications in daily life and industry

1. What is IoT?

Opening discussion

❖ **Which of the following can be considered IoT?**

- Smart watch
- Traffic camera
- Factory robot
- Delivery tracking system
- Air conditioner controlled by smartphone

Answers → Part 3

1. What is IoT?

Definition of IoT & important idea

- **Internet-of-Things** (IoT) **connects physical world** with **digital world**

- **IoT** is not only about **connecting devices**
- **IoT** is about **using data from physical things**
  - What is happening now?
  - Is the condition normal or abnormal?
  - Should the system send an alert?
  - Should an action be taken automatically?

1. What is IoT?

Examples of IoT in daily life

- **Smart watch** measuring heart rate
- **Smart home** – light, temperature control
- **Smart parking system –** obtain parking lot availability
- **Smartphone-connected air purifier** – control air freshness
- **Package tracking system** – track package location and arrival time

1. What is IoT?

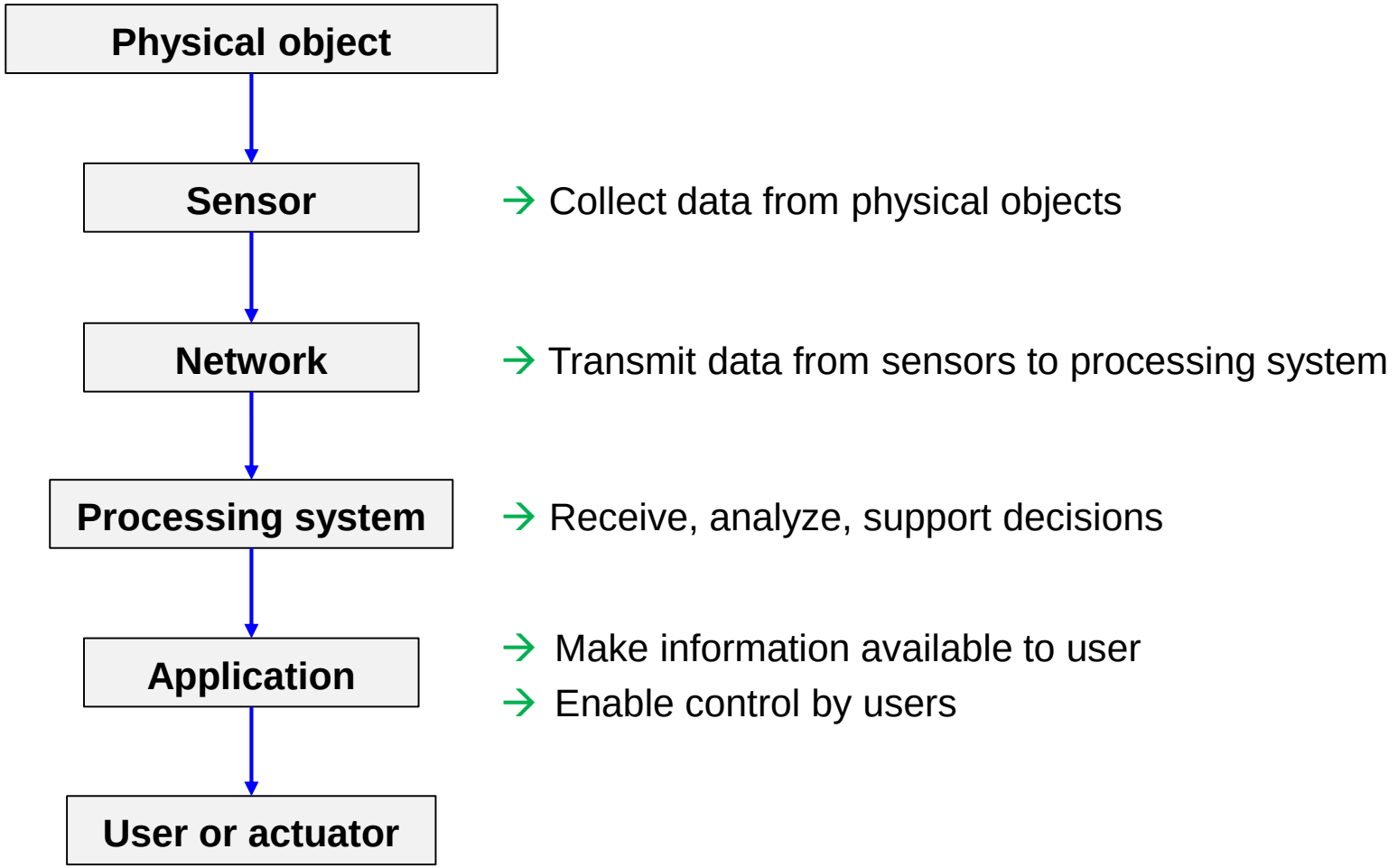
Examples of IoT in industry

- **Smart logistics** – obtain status of picking vehicles, empty lot, picking location
- **Automated inspection** – scan, inspect, feedback results
- **Energy monitoring** – obtain status, and remote control
- **Smart agriculture** – control temperature, humidity of farm

1. What is IoT?

Main components of IoT

❖ **IoT → a connected system, not only a single device**



1. What is IoT?

Sensors

❖ **Sensor → measure and obtain physical data from device**

Sensor type	Data
Temperature sensor	Heat
Humidity sensor	Moisture
Light sensor	Brightness
Camera	Image / video
Accelerometer	Linear motion
Gyroscope	Rotation and angular velocity
GPS	Location

1. What is IoT?

IoT Devices

- **An IoT device usually contains:**
  - Sensor
  - Processor
  - Memory
  - Communication module
  - Power source
  - Software
- **Example devices:**
  - Smart watch – sensors, Bluetooth, memory, processing, power source
  - Machine equipped with sensor
  - Vehicle tracking device – GPS, processing, communication, power management

1. What is IoT?

Networks in IoT

- **IoT devices** need **communication** to send data, receive commands
- Example networks:
  - Wi-Fi
  - Bluetooth
  - Ethernet
- **Network selection** depends on:
  - Distance
  - Speed
  - Power consumption
  - Cost
  - Reliability

1. What is IoT?

Processing systems

❖ **IoT data can be processed in different places**

Device



- ❖ Collects raw data
- ❖ Very limited CPU/memory
- ❖ Simple filtering
- ❖ Low latency at source

Read temperature
Check battery

Edge



- ❖ Processes data near devices
- ❖ Event detection
- ❖ Aggregation / filtering
- ❖ Reduced bandwidth and faster response

Detect abnormal
Trigger local alert

Cloud



- ❖ Large-scale storage
- ❖ Advanced analytics
- ❖ Model training / historical analysis
- ❖ Centralized management

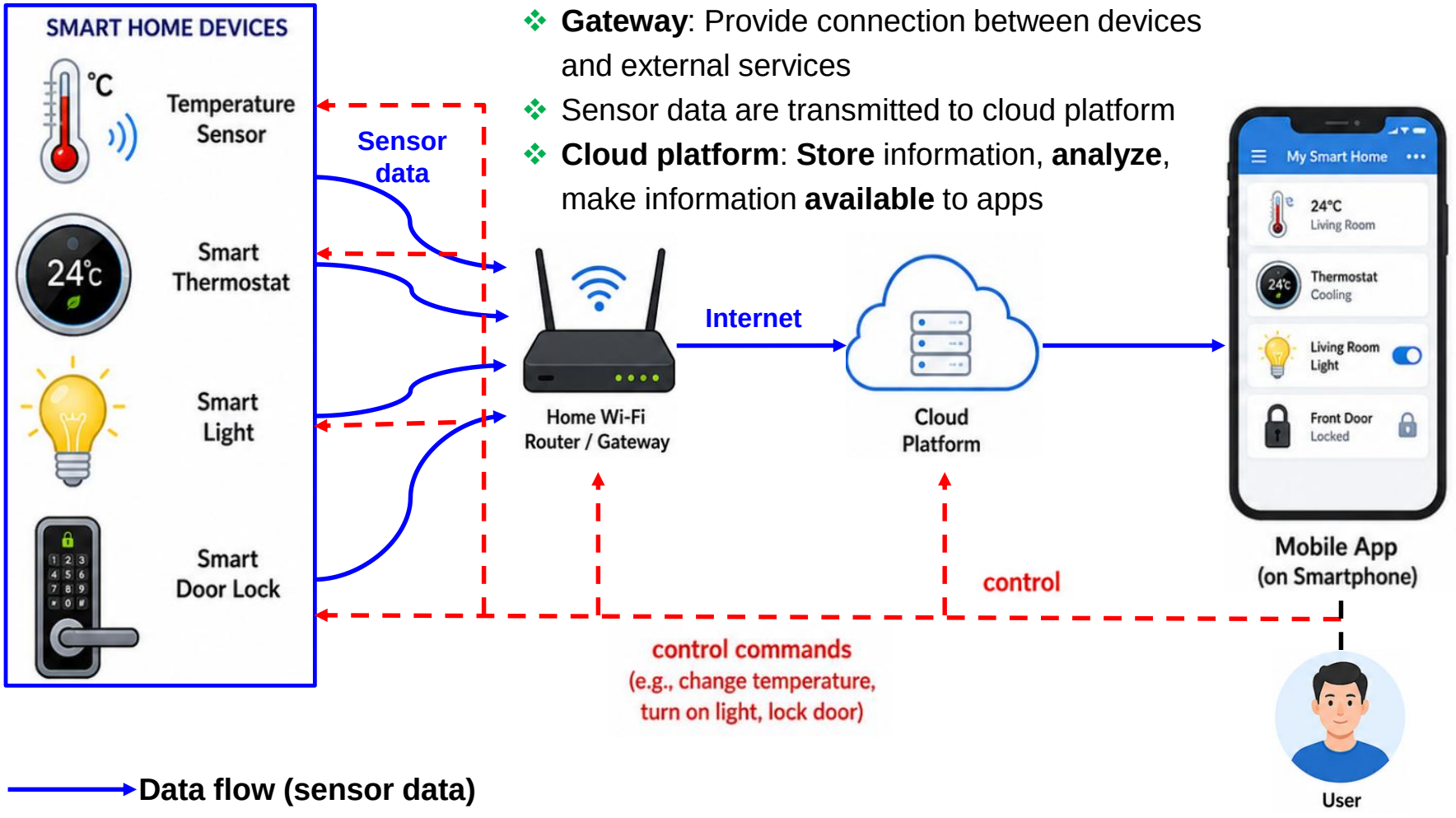
Store history
Analyze trends
Build dashboards

Key idea:

Device – handles **sensing; Edge** – handles **nearby processing; Cloud** – handles **large-scale storage and analytics**

1. What is IoT?

IoT in smart home



- **Gateway**: Provide connection between devices and external services
- Sensor data are transmitted to cloud platform
- **Cloud platform**: **Store** information, **analyze**, make information **available** to apps

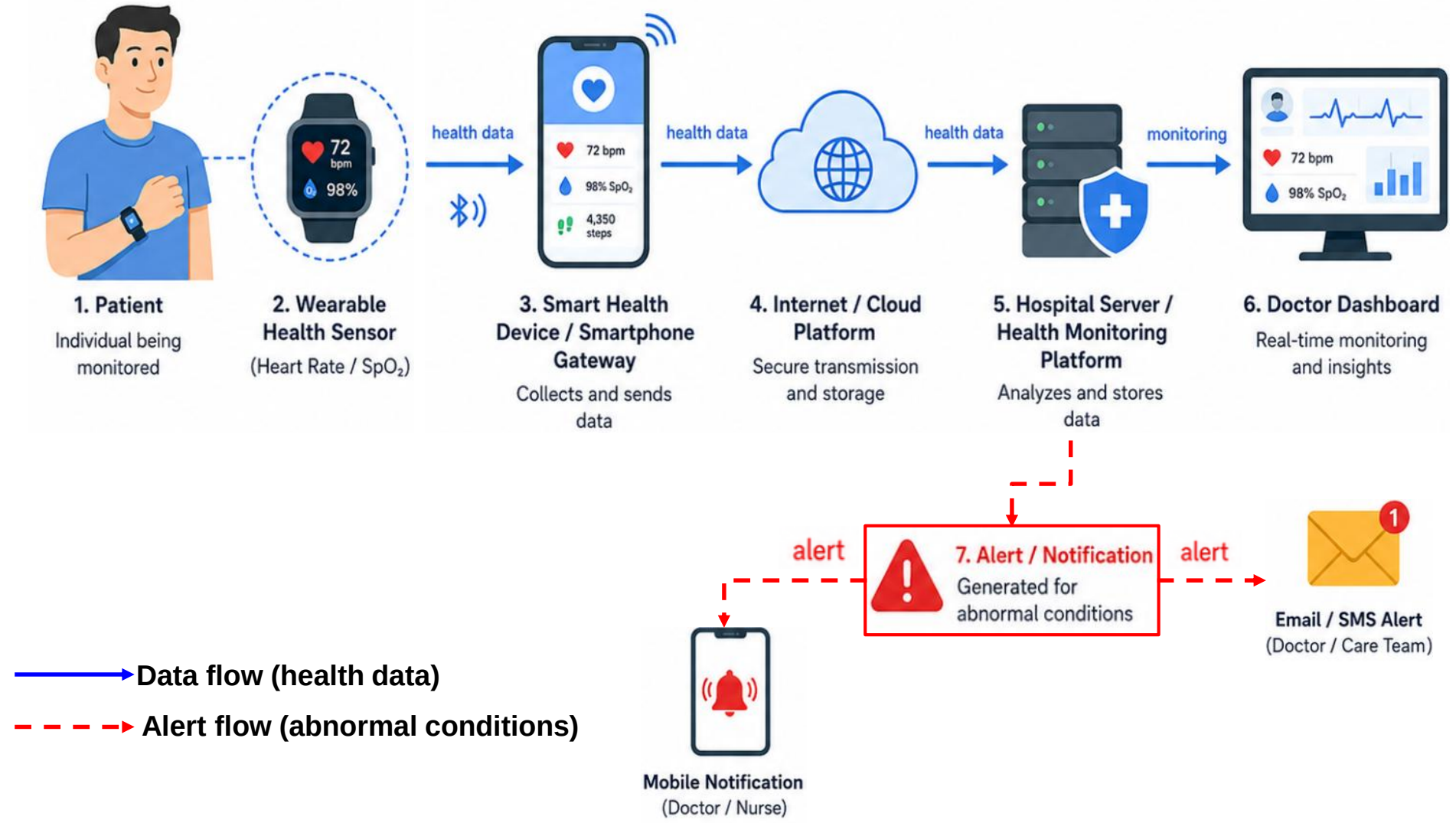
Data flow (sensor data)

Control flow (commands)

Monitor and **control** home devices via application

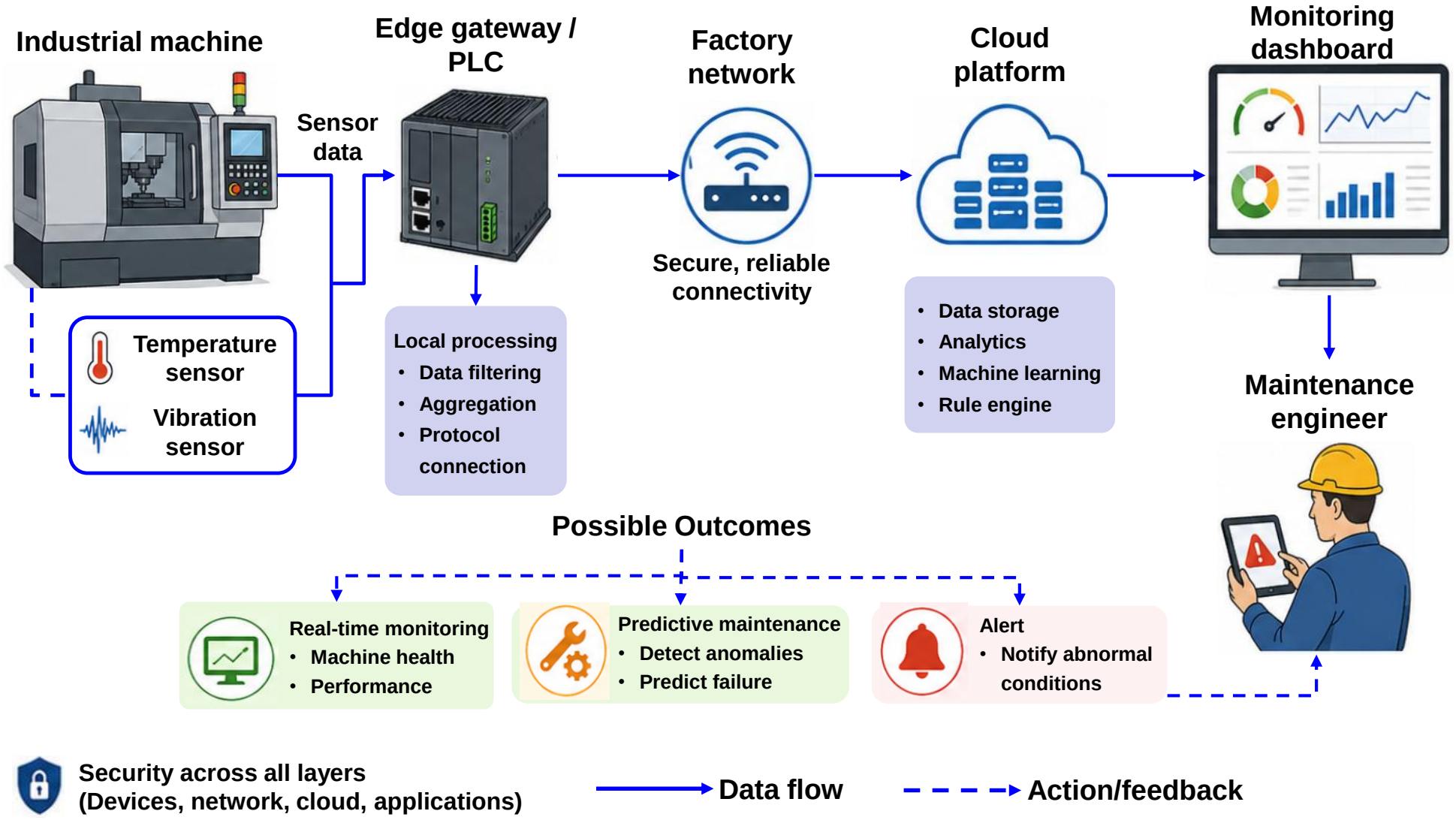
1. What is IoT?

IoT in healthcare



1. What is IoT?

IoT in machine monitoring



1. What is IoT?

Why IoT matters?

- **IoT is important** because it enables:
  - Real-time monitoring
  - Automation
  - Data-driven decision-making
  - Remote control
  - Predictive maintenance
  - Smart services

Part 1 Summary

- **IoT connects physical objects** to **digital systems**
- **IoT** systems **collect**, **transmit**, **process** and **use data**
- **Main components** include sensors, devices, networks, processing systems and applications
- **A device becomes part of IoT** when it can**:**
  - Sense or generate data
  - Communicate through a network
  - Interact with other digital system or application
- **Applications** of IoT: homes, cities, factories, healthcare, logistics and transportation

2. Understanding IoT systems

Learning objectives

By the end of this session, students should be able to:

- Understand how IoT systems differ from traditional computing systems
- Explain major characteristics of IoT systems
- Identify major IoT design challenges
- Explain why scalability, interoperability, reliability, and security matter

2. Understanding IoT systems

IoT system

A smart device alone does not represent the complete IoT system

- Example:

  Smart watch → Smartphone → Internet → Cloud platform → Health application → User

- **Complete IoT system** include
  - Physical device
  - Data
  - Communication
  - Processing
  - Application
  - User or action

2. Understanding IoT systems

Physical-digital loop

❖ **Many IoT systems operate as a continuous loop**

Physical Environment

↓

Sense

↓

Data

↓

Process

↓

Decision

↓

Action

↓

Physical Environment

Example:

Temperature rises

↓

Sensor detects 30°C

↓

System decides room is too hot

↓

Air conditioner turns on

↓

Temperature decreases

IoT systems can both**:**

- ❖ **Observe** the physical world
- ❖ **Control** the physical world

2. Understanding IoT systems

Traditional system vs IoT system

Traditional system	IoT system
Human enters data	Sensors generate data
Mostly digital environment	Physical + digital
Usually a few computing devices	Potentially many devices
User initiates many actions	Systems may act automatically
Computing often centralized	Computing may be distributed

❖ **Key differences:**

- Physical-world **interaction**
- **Large** number of devices
- **Continuous** data
- **Heterogeneous** hardware
- **Limited** computing, storage, communication resources
- **Distributed** computing

2. Understanding IoT systems

Characteristics 1: Many connected devices

- **IoT systems can range from:**
  - 5 sensors in a room
  - 500 sensors in a building
  - 10,000 devices in a factory
  - Millions of devices in a smart city

- **Possible problems when number of devices increases?**
  - More data
  - More communication
  - More management
  - More failures

2. Understanding IoT systems

Scalability

- **Scalability** → can the system **continue to work efficiently** as the **number of devices** or **amount of data increases**?
- Example:
  - 10 devices → easy to manage
  - 1,000 devices → more difficult
  - 1,000,000 devices → major system challenge
- **IoT platforms must scale in terms of:**
  - Devices
  - Messages
  - Storage
  - Processing
  - Users

2. Understanding IoT systems

Characteristics 2: Continuous operation

- **IoT devices** often **operate continuously**
- Example:
  - 1 sensor →1 reading/second = 86,400 readings per day
  - 1,000 sensors → 86,400,000 readings per day

Main issue:

Do we really need to send and store every measurement?

- **IoT data → changes over time**
- Example:
  - 10:00 → 23.5°C
  - 10:01 → 24.1°C
  - 10:02 → 24.8°C
  - 10:03 → 29.7°C

Main issue:

When was the value measured?

Is the value changing?

Is the current value abnormal?

Is historical information important?

2. Understanding IoT systems

Characteristic 3: Heterogeneity

- **IoT systems contain different types of devices**
- **Differences may include:**
  - Manufacturers
  - Sensors
  - Operating systems
  - Data formats
  - Communication technologies
  - Processing capabilities
- **Examples:**
  - Smart watch → small, battery-powered
  - Security camera → high data volume
  - Factory machine → high reliability required
  - Agricultural sensor → long-distance communication

Different IoT applications have different requirements

2. Understanding IoT systems

Different Applications Have Different Priorities

- IoT architecture is **application-dependent**

Application	Important requirement
Smart home	Convenience, cost
Healthcare	Reliability, privacy
Smart factory	Fast response, reliability
Smart agriculture	Long range, low power
Smart city	Scalability

2. Understanding IoT systems

Characteristic 4: Limited device resources

❖ **Many IoT devices have limited:**

- CPU
- Memory
- Storage
- Battery
- Communication capacity

❖ **Example comparison**:

Smartphone/PC	Small IoT Sensor
Powerful CPU	Small processor
GB of memory	KB/MB of memory
Rechargeable frequently	May operate for years
Fast network	Low-bandwidth network

2. Understanding IoT systems

Energy consumption

- For many IoT devices**, energy** is a major design issue
- **Energy is consumed by:**
  - Sensor measurement
  - Processing
  - Communication
  - Keeping network connection active
- **Important principle → Sending data can consume significant energy**

More communication

↓

More energy use

↓

Shorter battery life

2. Understanding IoT systems

Characteristic 5: Distributed systems

- **IoT processing does not necessarily happen in one place**

  Device → **nearby processing** → **remote processing**

- **Different parts** of the system may **perform different tasks**

  Where should processing happen?

- **Example: Machine temperature monitoring**

Task	Possible location
Read temperature	Device
Detect overheating	Device / Edge
Store one year of history	Cloud
Analyze factory-wide trends	Cloud
Trigger immediate alarm	Device / Edge

Processing location depends on **system requirements**

2. Understanding IoT systems

Challenge 1: Latency

- **Latency = delay between an event and the system response**
- **Example:**

Machine overheats

↓

Data sent to cloud

↓

Cloud processes data

↓

Command returned

What if this takes several seconds?

For safety-critical applications, faster local processing may be required

2. Understanding IoT systems

Challenge 2: Bandwidth

- Imagine: 1,000 sensors × 1 measurement every seconds
- **Sending everything to the cloud creates:**
  - High network traffic
  - Higher cost
  - More storage
  - More processing

Possible solution:

Raw data

↓

Local filtering

↓

Send important data only to cloud

2. Understanding IoT systems

Challenge 3: Network Dependency

What happens if the Internet fails?

- Possible consequences:
  - Data cannot be uploaded
  - Remote monitoring stops
  - Decisions may be delayed

IoT systems may therefore **need:**

- Local processing
- Local storage
- Retry mechanisms

2. Understanding IoT systems

Challenge 4: Reliability

- IoT systems may experience:
  - Sensor failure → health check / backup sensor
  - Communication failure → retry / local buffering
  - Gateway failure → backup gateway / local operation
  - Low battery → battery monitoring / power saving
  - Software errors → restart / error handling / update
  - Cloud failure → local processing / later synchronization

Should one failed component stop the entire system?

2. Understanding IoT systems

Challenge 5: Security

- IoT creates many possible attack points:
- **Possible risks**:
  - Unauthorized access
  - Fake sensor data
  - Data theft
  - Device takeover
  - Privacy leakage

2. Understanding IoT systems

Challenge 6: Privacy

- **IoT may collect**:
  - Location
  - Health information
  - Home activities
  - Video
  - Industrial operation data

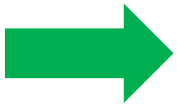
Does all collected data need to leave the local environment?

Edge computing

(Second half of this course)

Part 2 Summary

- **IoT systems** are **challenging** because they involve:
  - Many devices
  - Continuous data
  - Heterogeneous technologies
  - Limited resources
  - Distributed processing
  - Network dependency
  - Reliability
  - Security and privacy



- **IoT architecture must consider:**
  - Where to process?
  - What to send?
  - What to store?
  - How quickly must we respond?
  - How do we keep the system reliable and secure?

3. Practical

Practical learning objectives

By the end of the session, students should be able to:

- Decide whether a system can be considered IoT
- Trace how data moves through an IoT system
- Distinguish sensing, communication, processing, and action
- Design a simple conceptual IoT system

3. Practical

Activity 1: Is it IoT?

❖ **Which of the following can be considered IoT?**

- Smart watch
- Traffic camera
- Factory robot
- Delivery tracking system
- Air conditioner controlled by smartphone

3. Practical

Activity 1: Suggested answer

- **Potentially all of them,** if they are **connected** and **exchange data with other systems**

Example	Why it may be IoT
Smart watch	Sends sensor **data** to **smartphone/cloud**
Smart TV	**Communicates** with internet services
Traffic camera	Sends traffic **data to monitoring system**
Factory robot	**Exchanges operational data**
Delivery tracker	Sends **location/status** information
Smart air conditioner	**Receives commands** from smartphone

3. Practical

Activity 2: What does IoT data look like?

- **Consider a temperature monitoring device**

  Device ID: sensor_01

  Time: 10:32:15

  Temperature: 25.4°C

- **Identify:**

1. Device identity
2. Measurement time
3. Measured quantity
4. Measured value
5. Unit

Why Do We Need These Data Fields?

- **Device ID:** sensor_01
  - **Which device** produced the data?
- **Timestamp:** Time: 10:32:15
  - **When** was the measurement collected?
- **Sensor value:** Temperature: 25.4°C
  - **What was measured**?

3. Practical

IoT Data Changes Over Time

❖ **A sensor usually produces repeated measurements**

10:00 sensor_01 23.5℃

10:01 sensor_01 24.1℃

10:02 sensor_01 24.8℃

10:03 sensor_01 29.7℃

10:04 sensor_01 25.2℃

❖ **Questions:**

- Which reading is highest?
- Why is the timestamp important?
- Why might we want to keep previous readings?

3. Practical

Activity 3 — Trace an IoT System

❖ **Scenario: Smart home scenario**

- A temperature sensor measures room temperature
- The data is sent through Wi-Fi to a smart home application
- The user can view the temperature and control the air conditioner

❖ **Identify:**

❖ **Data flow**

Temperature
↓
Sensor
↓
Network
↓
Application
↓
User

❖ **Action/Control flow**

User
↓
Application
↓
Network
↓
Air Conditioner

IoT systems can both observe and control the physical world

3. Practical

Activity 4 — Arrange the components

Put these into a logical data flow

- **Smart health-monitoring system**
  - Doctor Dashboard
  - Wearable Sensor
  - Patient
  - Cloud Platform
  - Smartphone
  - Internet
- **Questions:**
  - Where is the physical world?
  - Where does sensing occur?
  - Where is communication used?
  - Who uses the information?

Suggested answer

- **Smart health-monitoring system**

Patient

↓

Wearable Sensor

↓

Smartphone

↓

Internet

↓

Cloud Platform

↓

Doctor Dashboard

3. Practical

Activity 5 — Smart Home, Healthcare and Industry

Complete the table (Given 3-4 minutes)

	Smart Home	Healthcare	Industry
Physical object	?	?	?
Sensor	?	?	?
Data	?	?	?
User	?	?	?
Possible action	?	?	?

3. Practical

Activity 5 — Smart Home, Healthcare and Industry

Possible answer

	Smart Home	Healthcare	Industry
Physical object	Room/home	Patient	Machine
Sensor	Temperature	Heart-rate sensor	Vibration sensor
Data	Temperature	Heart rate	Vibration
User	Resident	Doctor	Engineer
Action	Control AC	Send alert	Maintenance

The applications are different,

but the basic structure is similar

3. Practical

Activity 6 — Design your own IoT system

- **Example:** Smart classroom
- **Problem:** Classroom becomes too hot
- **System:**

  Room Temperature → Temperature Sensor → IoT Device → Wi-Fi → Monitoring System → Teacher / Building Manager → Air Conditioner

- **Data:**

  Device: classroom_sensor_01

  Time: 14:20

  Temperature: 29.5°C

- **Display on application:**

  Classroom A

  Temperature: 29.5°C

3. Practical

Activity 6 — Design your own IoT system

- Group discussion – 3 to 5 members
- Choose one application:
  - Smart home
  - Smart farm
  - Smart parking
  - Smart factory
  - Smart logistics
- Questions:
  1. What problem are you solving?
  2. What Should Be Measured?
  3. Build the System

3. Practical

Practice Roadmap

Week	Practical topics
1	Understand IoT systems without programming
2	Generate and process sensor measurements
3	Represent IoT data
4 – 6	Connect and communicate
7 – 15	Build increasingly capable IoT and edge systems

3. Practical

Preparation for Week 2

- Before the next practice class:
  - ✓ Form groups (3 to 4 members)
  - ✓ Bring a laptop if possible
  - ✓ Install Python 3.+
  - ✓ Prepare a Python editor
  - ✓ Check whether you can run:

```
print("Hello IoT")
```